



The Sarrus Oracle

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Difficulty Level : Medium

The Sarrus Oracle

In the old academy of arithmetic, a broken oracle is still used to test if a number is prime.

The theory behind it starts with Fermat's little theorem:

$$a^p \equiv a \pmod{p}$$

for every prime number p and every integer a .

This oracle uses the particular choice $a = 2$, so it checks whether a number satisfies the congruence

$$2^n \equiv 2 \pmod{n}.$$

The oracle incorrectly assumes the converse is also true: if a number satisfies

$$2^n \equiv 2 \pmod{n},$$

then it must be prime. But this is false: some composite integers still satisfy the same congruence.

In this problem, any composite integer with that property is called a **Sarrus number**. The classic example is $341 = 11 \times 31$.

So there are three important facts to keep in mind:

- every prime number passes the test
- some composite numbers also pass the test
- only those composite false positives are counted in this problem

That last point matters: prime numbers satisfy the congruence too, but they are **not** Sarrus numbers and must not be counted.

The academy stores candidate numbers in long consecutive archives. For each interval $[L, R]$, determine how many integers in that interval would fool the oracle – in other words, how many Sarrus numbers belong to that range.

Task: For each query $[L, R]$, count the composite integers n such that $L \leq n \leq R$ and $2^n \equiv 2 \pmod{n}$.

You may think of each query as one question asking:

“Inside the interval $[L, R]$, how many composite numbers look prime to the oracle?”

Input

The first line contains a single integer q ($1 \leq q \leq 2 \cdot 10^5$) – the number of queries.

Each of the next q lines contains two integers L and R ($2 \leq L \leq R \leq 2 \cdot 10^6$), describing one interval.

Output

For each query, print a single integer: the number of Sarrus numbers in the interval $[L, R]$.

Constraints

- $1 \leq q \leq 2 \cdot 10^5$
- $2 \leq L \leq R \leq 2 \cdot 10^6$
- Time limit: **3 seconds**
- Memory limit: **512 MB**

Sample Input 1

```
4
2 100
300 400
500 700
1000 2000
```

Sample Output 1

```
0
1
2
4
```

Explanation

- There are no Sarrus numbers between 2 and 100.
- In $[300, 400]$, the only one is 341.
- In $[500, 700]$, the two Sarrus numbers are 561 and 645.
- In $[1000, 2000]$, they are 1105, 1387, 1729, and 1905.

Notice that prime numbers inside those intervals are ignored, even though they also satisfy the oracle’s congruence.